StepFun Part II: Farseer's loss surface and allocation experiments

Interaction terms, differential fitting, two-dimensional designs and out-of-sample prediction

What Part II adds

Li et al., Farseer v3 studies prediction of final model loss across parameter count and token horizon, after standardizing the training family and choosing suitable HPs. It is distinct from Step Law's optimizer-choice surface.

Primary local artifacts: PDF, LaTeX/code metadata, and the archived official implementation. We have reviewed the method and source; we have not rerun the approximately 1,000-model suite.

1. Define a family, not just a parameter count

AppendixA defines an architecturally standardized family: N must map to width, depth, attention, FFN, normalization and other choices. Otherwise two equal-N points can represent different algorithms. A dataset recipe similarly includes the tokenizer, processing and mixture policy, not merely its name.

The reported setup (§2.2) uses web/code/math mixtures, BPE vocabulary 65,536, context 2,048, AdamW $(0.9, 0.95)$, epsilon10^{-8}, decay 0.1, clip 1, 2,000 warmup updates and cosine decay to absolute LR10^{-5}. Sizes and horizons are geometrically spaced; a separate high-quality validation set contains30M tokens. AppendixA uses Step Law-style settings to limit suboptimal-HP noise.

The primary compression metric is **bits per character**, not Marin's bits per byte. With non-ASCII text the denominators differ. Keep metric/text normalization/boundary conventions fixed when comparing fits or mixtures.

2. Why add a size–data interaction?

A separable Chinchilla-style model is

$$L(N, D) = E + AN^{-\alpha} + BD^{-\beta}.$$

At fixed N, all sizes share one data exponent β; only the additive model term changes. Farseer instead starts with

$$L(N, D) = E + U(N) + V(D) + H(N, D),$$

where H allows the response to more data to depend on model size. This is an empirical modeling hypothesis, not proof that a fitted interaction has a unique mechanistic interpretation.

For a data multiplier $r > 1$, a difference at fixed N removes the size-only term:

$$\Delta_D L(N, D) = L(N, D) - L(N, rD).$$

If the data-dependent component is $B(N)D^{-A(N)}$, then

$$\Delta_D L(N, D) = B(N)(1 - r^{-A(N)})D^{-A(N)}.$$

This explains the differential analysis: the log-log slope of those differences can diagnose a size-dependent data exponent. Differences amplify measurement noise and correlated endpoints, so uncertainty cannot be ignored.

3. The fitted functional form

The paper's Eq.3 is

$$\widehat{L}(N, D) = e^{a_3 N^{\gamma} + b_3} + e^{a_2 N^{\beta} + b_2} D^{-e^{a_1 N^{\alpha} + b_1}}.$$

Equivalently, write $R(N) + B(N)D^{-A(N)}$, with positive A, B, R parameterized by exponential functions. There are nine coefficients/exponents; their values and numerical units are suite-specific. A flexible expression with more coefficients is not automatically better outside the fitted domain.

The method uses differential/piecewise fits to estimate data-dependent slopes/amplitudes, examines their dependence on N, estimates the remaining size-dependent residual, then refines fits with the reported iterative procedure. Read the official code/AppendixC for exact objective weighting, transformations and constraints. Directly launching one unconstrained nonlinear optimizer from arbitrary starts need not reproduce the paper's estimates.

4. What data geometry is needed?

A fixed-TPP ladder has $D = rN$, so it samples only one ray through (N, D). It may characterize performance along that recipe but cannot independently identify a general two-dimensional allocation surface.

For Farseer-style analysis, train multiple horizons per size and multiple sizes over overlapping horizon ranges. Hold out entire sizes and horizon ranges; randomly holding out nearby points on the same training trajectory overstates extrapolation evidence. A shared-prefix checkpoint is not a fully annealed endpoint for a shorter horizon unless the appropriate annealing branch was actually trained.

The paper analyzes robustness to fit-data coverage, a different English/Chinese recipe and extrapolation to larger models (§4). Those are empirical tests on its suite. They do not establish calibrated error bars for new sparse/recurrent models, new schedules or cyber data.

5. From a surface to a compute allocation

For a dense approximate convention $C = 6ND$, the fitted allocation problem is

$$N_{\star}(C) = \arg\min_{N} \widehat{L}\left(N, \frac{C}{6N}\right).$$

For the separable baseline, the stationary equation is

$$\alpha A N^{-\alpha} = \beta B (6N/C)^{\beta}.$$

It produces power-law allocations under the model assumptions. Farseer's size-dependent data exponent generally requires a numerical optimization over the feasible family. Do not infer a universal20TPP or a universal increasing TPP from an empirical fit whose family/metric changed.

For MoE, use measured/derived architecture FLOPs $C = f(\mathcal{A}, \ell)D$ and maintain stored/active parameter definitions. A size label alone does not fix cost when expert count, top-k, latent dimension or recurrence changes. Serving constraints can favor an intentionally overtrained smaller model even when the training-loss allocation optimum differs.

6. A defensible fitting and comparison workflow

1. Tune the recipe adequately first; retain its uncertainty and failure regimes. A bad LR at one horizon can masquerade as a new scaling exponent.
2. Freeze preprocessing, data mixture, tokenizer, family mapping and evaluation text. Compare different families on a common target metric.
3. Fit the simpler separable law and Farseer on identical fitting points, with multiple initializations/appropriate constraints and declared weights.
4. Inspect residuals by width, horizon, TPP and schedule phase. Report absolute and relative held-out errors, not only in-sample R^2.
5. Bootstrap independent runs/cells, account for shared prefixes, and stress-test sensitivity to dropping a size or longest horizon.
6. Predict an untouched larger/longer run before inspecting its outcome. If the forecast misses, diagnose the recipe and data transfer, not only the curve fit.
7. Compare recipe families at both matched analytical compute and wall time. A lower fitted FLOP frontier can lose after communication or optimizer overhead.

When should our team run this?

Do it if model/data allocation or a calibrated hero-loss forecast materially affects the decision. For a constrained96-H100 run with a serving-size target and a measured corpus, it is reasonable to prioritize the HP ladder and fresh long-horizon validation first. If we skip the loss-surface study, say that we have **no calibrated hero-loss prediction**.

For cyber data, first test a controlled mixture addition on held-out domain/general metrics. If we later compare scaling surfaces, freeze each data recipe and fit its own surface with matched HP-tuning effort. A single small-model improvement does not establish a hero-scale advantage.

See Marin's search/loss-law sequence and the current resource plan for the practical decision gates.